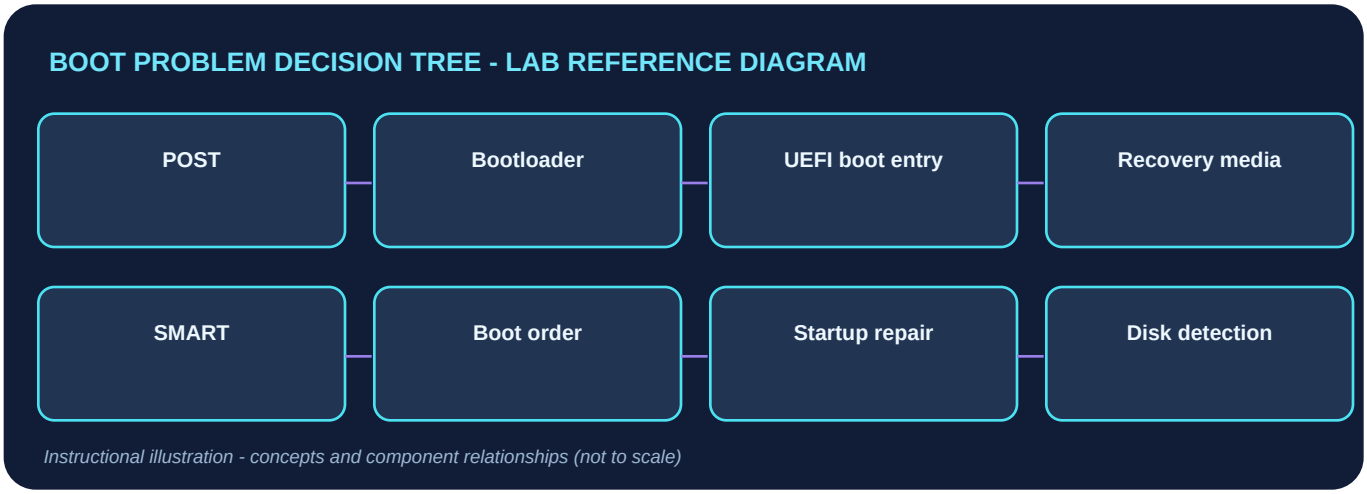
4.2.6 Troubleshoot Boot Issues

STUDENT STUDY GUIDE + VOCABULARY + VISUAL REFERENCE

Learning goal

Diagnose failed startup conditions.

Reference image



Use the labeled illustration to identify the key components, commands, or checkpoints you will encounter in this lab.

Prepare for the simulation

1. Identify whether failure occurs before or after POST.
2. Check whether firmware detects the boot device.
3. Confirm the correct boot mode and order.
4. Use recovery tools only after safeguarding data.

Troubleshooting and safe practice

• No bootable device: check disk detection and UEFI entry.

• Repeated restart: review diagnostics and startup logs.

4.2.6 | VOCABULARY & REVIEW

Essential vocabulary

Term	Meaning / recognition cue
POST	Power-on self-test; firmware checks essential hardware during startup.
Bootloader	Software that starts the operating system after firmware.
UEFI boot entry	A key term in boot problem decision tree; identify its purpose, where it is used, and how to verify it.
Recovery media	A key term in boot problem decision tree; identify its purpose, where it is used, and how to verify it.
SMART	Storage drive health monitoring and diagnostic information.
Boot order	Priority in which firmware tries eligible boot devices.
Startup repair	A key term in boot problem decision tree; identify its purpose, where it is used, and how to verify it.
Disk detection	Storage device or management concept; confirm detection and preserve existing data.

Knowledge check - answer in your own words

1. Identify two components or commands in the reference diagram and explain what each does.

2. What should you verify before changing or installing hardware, settings, or files in this lab?

3. Choose one listed failure symptom. What is your first safe diagnostic step?

4. How will you prove the task is complete, rather than merely assuming success?

Lab readiness checklist

[ ] I can identify the labeled reference items. [ ] I understand the safety precautions.

[ ] I can explain the vocabulary. [ ] I can verify and document successful results.

Student: _____ Date: _____